Yunfan He

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Work

Postdoctorial Fellow
Beijing International Center for Mathematical Research *Sep 2024 - Present*
Mentor: Bohan Fang

Education

Ph.D. Mathematics
University of Wisconsin-Madison *2018 – 2024*
Advisor: Andrei Căldăraru

M.A. Mathematics
University of Wisconsin-Madison *2016 – 2018*

B.S. Mathematics and Applied Mathematics
Shanghai Jiao Tong University *2013 – 2017*

Experience

Graduate Student Teaching Assistant
University of Wisconsin-Madison *Aug 2018 – May 2024*

Awards, Honors and Grants

The Excellence in Research Graduate Student Awards
University of Wisconsin-Madison *2023,2024*

Summer Research Fellowship
University of Wisconsin-Madison *2020,2021,2023*

Outstanding TA Award
University of Wisconsin-Madison *2020*

Publications and Preprints

[1] Andrei Caldararu and Yunfan He. *Holomorphic anomaly equation for categorical enumerative invariants of elliptic curves.* (in preparation).

[2] Andrei Caldararu, Yunfan He, and Weng-Him Cheung. *g=2 categorical enumerative invariants for elliptic curves.* (in preparation).

[3] Andrei Caldararu, Yunfan He, and Shengyuan Huang. *Moonshine at Landau-Ginzburg points*. *Proceedings of the American Mathematical Society*. arXiv: 2107.12405. (published online).

[4] Yunfan He. "Categorical Enumerative Invariants of Elliptic Curves". PhD thesis. The University of Wisconsin-Madison, 2024.

[5] Yunfan He. "Hodge-to-de Rham degeneration of nodal curves". In: *Transactions of the American Mathematical Society* 379.12 (2026), pp. 8965–8989. arXiv: 2311.08477.

[6] Yunfan He and Junxiao Wang. *Derived Hodge-to-de Rham Degeneration and Quasihomogeneous Curve Singularities*. 2026. arXiv: 2604.05825v2. (submitted).

Conference Talks

HAE for CEI of elliptic curves is "easy".
Simons Center for Geometry and Physics *February 2025*
Recent developments in higher genus curve counting

HAE for categorical enumerative invaraints of elliptic curves.
University of Arizona *May 2024*
Commutative Algebra and Algebraic Geometry in TUcSon

HAE for categorical enumerative invaraints of elliptic curves.
University of Oregon *April 2024*
Workshop on Enumerative Geometry and related topics

Seminar Talks

Hodge-to-de Rham Degeneration for Singular Curves.
Peking University *May 2026*
Postdoc Seminar

HAE for categorical enumerative invariants.
Peking University *November 2024*
Postdoc Seminar

Hodge to de Rham degeneration of nodal cubic curve.
Simons Laufer Mathematical Sciences Institute *Spring 2024*
COMA/NAG Graduate Student Seminar

Variation of Hodge structures.
University of Wisconsin-Madison *Spring 2023*
Graduate Algebraic Geometry Seminar

Introduction to mixed Hodge structures and mixed Hodge modules.
University of Wisconsin-Madison *Fall 2020*
Graduate Algebraic Geometry Seminar

Braid group action on derived categories.
University of Wisconsin-Madison *Spring 2020*
Graduate Algebraic Geometry Seminar

Finite common covering of graphs.
Shanghai Jiao Tong University *Summer 2016*
Combinatorics Seminar

Conferences and Workshops

2025 International Congress of Basic Science
Beijing Institute of Mathematical Sciences and Applications *July 2025*
Participant

String-Math 2025
Beijing Institute of Mathematical Sciences and Applications *June 2025*
Participant

Recent developments in higher genus curve counting
Simons Center for Geometry and Physics *Februay 2025*
Speaker

Topological Strings on Calabi-Yau Manifolds - Math and Physics
Beijing International Center for Mathematical Research *September 2024*
Participant

Commutative Algebra and Algebraic Geometry in TUcSon.
University of Arizona *May 2024*
Speaker

Workshop on Enumerative Geometry and related topics
University of Oregon *April 2024*
Speaker

Recent Developments in Noncommutative Algebraic Geometry
Simons Laufer Mathematical Sciences Institute *April 2024*
Participant

Introductory Workshop: Noncommutative Algebraic Geometry
Simons Laufer Mathematical Sciences Institute *February 2024*
Participant

Connections Workshop: Noncommutative Algebraic Geometry
Simons Laufer Mathematical Sciences Institute *February 2024*
Participant

FRG workshop on Brauer groups and derived categories
Northwestern University *October 2023*
Participant

Algebraic Geometry Northeastern Section
University of Pennsylvania *October 2023*
5 mins talk speaker

FRG preprint seminar
Northwestern University *August 2023*
Participant

Algebraic geometry and cohomology in mixed characteristic
Northwestern University *May 2023*
Participant

Midwest Algebraic Geometry Graduate Conference
University of Illinois-Chicago *August 2022*
Participant

Derived FRG: Derived Categories, Moduli Spaces, and Hyperkähler Varieties
University of Michigan-Ann Arbor *August 2022*
Participant

Algebraic Geometry Northeastern Series
Rutgers University(online) *May 2022*
Participant

On the crossroads of algebra, geometry and physics - Dedicated to A. Goncharov
Yale University *March 2022*
Participant

Algebraic Geometry Northeastern Series
Boston College(online) *October 2021*
Participant

Madison Moduli Weekend
University of Wisconsin-Madison(online) *September 2020*
Participant

Midwest Algebraic Geometry Graduate Conference
University of Illinois-Chicago(online) *May 2020*
Participant

Groups and Graphs, Designs and Dynamics(G2D2)
Three Gorges Mathematical Research Center *August 2019*
Participant

Midwest Algebraic Geometry Graduate Conference
University of Illinois-Chicago *March 2019*
Participant

I-70 Algebraic Geometry Symposium
University of Missouri-St. Louis *October 2018*
Participant

Teaching

Instructor at Peking University
- **Advanced Mathematics C** (*Instructor*) *Fall 2026*

Lecturer Student Assistant at University of Wisconsin-Madison
- **Math 112: College Algebra** (*Instructor*) *Fall 2023*

Graduate Student Teaching Assistant at University of Wisconsin-Madison
- **Math 321: Applied Mathematical Analysis** (*TA*) *Spring 2023*
- **Math 234: Calculus III** (*TA*) *Fall 2022*
- **Math 240: Introduction to Discrete Mathematics** (*TA*) *Summer 2022*
- **Math 340: Matrix and Linear Algebra** (*TA*) *Spring 2022*
- **Math 320: Linear Algebra and Differential Equations** (*TA*) *Fall 2021*
- **Math 234: Calculus III** (*TA*) *Spring 2021*
- **Math 320: Linear Algebra and Differential Equations** (*TA*) *Spring 2020-Fall 2020*
- **Math 112: College Algebra** (*TA*) *Fall 2019*
- **Math 222: Calculus II** (*TA*) *Spring 2019*
- **Math 221: Calculus I** (*TA*) *Fall 2018*

Outreach and Service

Graduate Student Mentor
University of Wisconsin-Madison *Spring 2018*
Directed Reading Program

Volunteered Tutor
Shanghai Jiao Tong University *Aug 2014*
Volunteer Tutor Program